

VIJAYA MALLIPUDI

Data Engineer · Python · SQL · PySpark · Databricks · Airflow · AWS · Banking & Healthcare
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github.com/vijayamallipudi/EnterpriseDocumentChatbot

SUMMARY

Data Engineer with experience across banking, healthcare, and SaaS. Build and operate pipelines on Python, SQL, PySpark, Databricks, Apache Airflow, and AWS processing ~2M records daily into validated, monitored datasets for risk, clinical, and product teams. Focused on pipelines that fail loudly rather than silently.

EXPERIENCE

Data Engineer · Fifth Third Bank

Jan 2025 – Present | Remote

Support the data layer behind credit risk reporting and analytics.

- Automated movement of loan and customer data across MySQL, PostgreSQL, and Redshift using Python and Apache Airflow, processing ~2M records daily and replacing manual reconciliation with scheduled, monitored workflows.
- Added schema validation and data quality gates to credit risk pipelines after a silent upstream type change corrupted downstream risk metrics pipelines now fail loudly on schema drift, row count deviation, and out-of-range business metrics rather than completing successfully on bad data.
- Configured Airflow alerting on execution failures and data anomalies, moving detection of data issues from downstream business users to the pipeline itself.
- Diagnosed a reporting discrepancy by tracing row counts and sample records across source, staging, and final layers; isolated the upstream schema change, corrected the PySpark transformation, backfilled affected partitions, and validated output against prior runs.
- Built and maintained 40+ curated datasets on Databricks and Redshift, giving 20+ risk and reporting stakeholders self-serve access to credit data and reducing dependency on ad-hoc requests.
- Tuned high-cost credit risk queries across Redshift, PostgreSQL, and MySQL - restructured joins, revised sort and distribution keys, and replaced nested subqueries with CTEs to shorten the daily reporting window.
- Built S3-to-Redshift ingestion pipelines feeding PySpark transformation workloads on Databricks for risk model inputs.

Stack: Python · SQL · PySpark · Databricks · Apache Airflow · AWS S3 · Amazon Redshift · MySQL · PostgreSQL · Git

Data Engineer · Elevance Health

Jun 2024 – Jan 2025 | Atlanta, GA

Kept claims and reporting data flowing for 15+ clinical and operational teams.

- Built SQL and PySpark pipelines processing ~1.5M healthcare claims records daily, handling extraction, cleaning, and standardization across source systems feeding clinical reporting.
- Built pipelines and queries on masked and de-identified claims datasets under role-based access controls, ensuring transformations preserved those protections downstream and complied with enterprise data governance policies.
- Migrated core transformation workflows to Databricks on AWS, consolidating fragmented processing logic and improving consistency across large claims datasets.
- Maintained 30+ reporting datasets at a 99%+ execution rate across 15+ teams, monitoring pipeline health and resolving failures before they reached downstream consumers.

- Delivered curated datasets through AWS Glue and S3, shortening the path from raw claims data to clinical team reporting.

Stack: Python · SQL · PySpark · Databricks · Apache Airflow · AWS S3 · Amazon Redshift · AWS Glue · PostgreSQL · MySQL · Git

Data Engineering Intern · ByteRidge

Jun 2021 – Aug 2023 | Hyderabad, India

Built and maintained pipelines spanning reporting, product, and internal operations.

- Designed and deployed Airflow DAGs from scratch across 18 ingestion and processing pipelines defining schedules, task dependencies, retry behavior, and failure notifications, and integrating the underlying ETL logic.
- Added monitoring and structured logging across orchestrated workflows, reducing time to diagnose pipeline failures.
- Developed Python workflows to clean, transform, and reconcile data across MongoDB and AWS EC2.
- Maintained 25+ operational datasets serving 10+ business and product users, standardizing recurring reporting that previously required manual preparation.

Stack: Python · SQL · Apache Airflow · MongoDB · AWS EC2 · AWS S3 · Git

PROJECT

Enterprise Document Chatbot (Local RAG) | Document Ingestion & Retrieval Pipeline

- Built a fully local document ingestion and retrieval pipeline using Python, LangChain, Ollama, ChromaDB, and FastAPI to process unstructured PDF/DOCX/TXT documents into searchable vector embeddings, with every answer grounded in and citing the source document.
- Designed the retrieval pipeline end to end: chunking, embedding, and RAG-based retrieval workflows, plus normalization/preprocessing logic to improve retrieval relevance.
- Developed a FastAPI backend with streaming, token-by-token responses, powered entirely by local LLM inference via Ollama — no OpenAI or Pinecone dependency, no API keys, no per-query cost.
- Made a deliberate architecture tradeoff to keep each Q&A stateless (no conversation history) after observing the local model lose context when prior turns were carried forward.

Stack: Python · LangChain · Ollama · ChromaDB · FastAPI · RAG · Vector Embeddings · Prompt Engineering

EDUCATION

M.S. Computer Science · University of Central Oklahoma

Aug 2023 – May 2025 | Edmond, OK

B.Tech Computer Science · Gayatri Vidya Parishad College of Engineering for Women · 2023

TECHNICAL SKILLS

Data Engineering: Data Pipelines, ETL/ELT, Workflow Orchestration, Data Modeling, Data Warehousing, Data Quality Validation, Schema Validation, Batch Processing

Technologies: Apache Airflow, Apache Spark, PySpark

Cloud & Storage: AWS S3, Amazon Redshift, AWS Glue, AWS EC2, Databricks, Snowflake

Programming: Python (Pandas, NumPy, PySpark), SQL (CTEs, Window Functions, Query Optimization)

Databases: PostgreSQL, MySQL, MongoDB

Tools: Git, GitHub, FastAPI

AI & Retrieval: RAG, LangChain, ChromaDB

Domain Experience: Banking, Healthcare, Credit Risk, Claims Data, Regulated Data Environments